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Planning Optimal Trajectory for Histogram-Enabled Mapping And Navigation by an Efficient PSO Algorithm

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***Abstract* — A particle swarm optimization (PSO) algorithm associated with a histogram navigation method is proposed in this paper for real-time map building and path planning for *multiple goals* applications. In real world applications such as rescue robots, service robots, mining mobile robots, and mine searching robots, etc., an autonomous mobile robot needs to reach multiple goals with a shortest path that, in this paper, is capable of being implemented by a PSO method with minimized overall distance. Once a global trajectory is planned, a foraging-enabled trail is created to guide the robot to the multiple goals. A histogram-based local navigation algorithm is employed to plan a collision-free path along the trail planned by the global path planner. A replanning-based algorithm aims to generate path while a mobile robot explores through a terrain with map building in unknown environments. In this paper, simulation and experimental results demonstrate that the real-time concurrent mapping and multi-goal navigation of an autonomous robot is successfully performed under *unknown* environments.**

I. Introduction

MOTION planning and mapping of autonomous robots is an important issue. Concurrent motion planning and mapping of robotics is to search a suitable collision-free path of an autonomous robot to move from an initial position to a goal designation while the robot builds up a map, in a unknown environment. Research on such point-to-point robot motion planning has been well carried out recently, indeed, which is a sort of single-goal navigation. In real-time world applications, such as rescue robots, service robots, mine rescue robots, and mine searching robots, etc., **multi-goal** motion planning of autonomous robots is highly desirable. In addition to the service robots, many other robotic applications require multi-goal navigation such as vacuum robots, land mine detectors, lawn mowers, and windows cleaners. The multi-goal path planning aims to search a collision-free path for visiting a sequence of *goals* with the minimized total route under unknown environments.

There have been a large number approaches proposed on autonomous robot motion planning ([1-7]). Gu *et al.* [1] proposed an elastic-band-based motion planning model, in which tunability and stability are focused on the robot navigation. Raja *et al.* [2] proposed an introducing a gradient function in the conventional potential field method associated with a GA-based method for robot motion planning. Besides an attractive force, a repulsive force, a tangential force and a gradient force are introduced in the conventional potential field method integrated with GA path planning model. Davies and Jnifene [3] developed a Genetic Algorithm (GA) path planner to guide an autonomous robot to reach specified multiple goals with obstacle avoidance. However, a couple of *artificial* waypoints should necessarily be predefined to assist in preventing the robot from the deadlock or escaping from the local minima, which causes more timing delay, more length and more cost.

Luo and Yang [4-5] developed a bio-inspired neural network model that concurrently performs mapping and path planning tasks. Yang and Meng [6] proposed a biologically inspired neural network approach to real-time collision-free motion planning of mobile robots in a non-stationary environment. Each neuron in the topologically organized neural network has only local connections. The real-time robot motion is planned through the dynamic activity landscape of the neural network. However, multi-goal navigation studies have not been performed largely for intelligent robot systems. Faigl and Macak [7] developed a self-organizing map method in conjunction with an artificial potential field based navigation function to generate an optimal path of a mobile robot to visit multiple goals, through a Traveling Salesman Problem (TSP) tool. The autonomous robot visits multiple targets just as the traveling salesman problem but with the presence of obstacles, in which mobile robots are navigated with the discrete map representation filled with some exact cell in workspace. Although a multi-goal navigation model of an autonomous robot modeled as a point robot is implemented by Gopalakrishnan, and Ramakrishnan [8], the model lacks of map building component, neither.

In this paper, a real-time concurrent multi-goal motion planning and map building approach of an autonomous robot in completely **unknown** environments is proposed. A local map composed of cells is dynamically built through the histogram-driven local navigator with restricted sensory information such as LIDAR-based data while it is navigated under completely unknown environments with obstacle avoidance to visit multiple requested goals. The sensory information obtained by onboard sensors mounted on the robot is utilized for its real-time concurrent multi-goal navigation and map building in unknown sceneries.

The rest of this paper is organized as follows. The well-known D*-Lite global path planner and a Modified VFH

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local navigator are addressed in Section II and III, respectively. In Section IV, the map representations of the environments and map building are introduced. The collision-free multi-goal navigation of the autonomous mobile robot by PSO-based TSP is presented in Section V. The real-time concurrent map building and collision-free multi-goal navigation of the autonomous robot are reported in Section VI. Afterwards, the simulation studies and experimental test of the real-time multi-goal navigation and map building of the autonomous robot in various unknown environments are described in Section VII. Finally, several important properties of the proposed approach are concluded in Section VIII.

II. HISTOGRAM -BASED LOCAL NAVIGATOR AND CELL-BASED MAP BUILDING

The navigation system consists of two layers, one is a D*-Lite global path planner with replanning function and the other is a histogram-based local navigator.

Flexibility and efficiency motivate D*-Lite to be adapted to multi-goal path planning. D*-Lite is an incremental heuristic search algorithm extended from A* algorithm by re-utilizing previous search effort in subsequent search iterations for efficient replanning. D* Lite is based on Lifelong Planning A* introduced by Koenig and Likhachev ([9]), which is an incremental replanning version of A*. A* utilizes a best-first search from a starting point ***S***_start to the goal ***S***_goal guided by the heuristic *h* thus it is able to search an efficiently traversable path between points commonly employed for robotics path planning in known 2D gird-based maps ([10]). The primary principle of D*-Lite in terms of incremental search is to re-utilize previous search effort in the subsequent search iterations, which represents an incremental heuristic replanning version of A*. The incremental heuristic search based D*-Lite algorithm is therefore orders of magnitude more than efficient than A* that plans a new path from scratch repeatedly. In comparison with A*, computational effort of D*-Lite with regard to search time may decrement by a factor of one to two orders of magnitude ([10]). In this paper, the D*-Lite is employed to generate the global trajectory of an autonomous robot under ***unknown*** environments.

The local navigator aims to generate velocity commands for the autonomous mobile robot to move towards a goal. The inclusion of a sequence of markers in the motion planning, which decomposes the global trajectory generated by the global planner into a sequence of segments, makes the model especially efficient for the workspace densely populated by obstacles. Ulrich and Borenstein [11] first successfully proposed a Vector Field Histogram (VFH) methodology for navigation. The Virtual Force Field (VFF) approach was initially inspired by potential field method in conjunction with the concept of certainty cells. Due largely to failure of avoiding the inherent defects of potential fields, its forward version, Vector Field Histogram (VFH) method was developed to overcome the shortcomings of VFF ([11]). Afterwards, the enhanced version of VFH approach, namely, VFH, was introduced to take the dimension of robot into account. Use of A* search algorithm associated with heuristics and appropriate cost function, the VFH was evolved to VFH*. In this paper, VFH is utilized as our LIDAR-based local navigator.

2D cell-based map filled with equally-sized cells, which are marked as either occupied or free, is built as the mobile robot moves. Concurrent map building and navigation are the essence of successful multi-goal navigation under unknown environments. Map building is a fundamental task in order to achieve high levels of autonomy and robustness in multi-goal navigation that makes it possible for autonomous robots to make decision in positioning with obstacle avoidance.

It is especially beneficial for autonomous robots to perform robust multi-goal navigation in unknown terrains, given the fact that it facilitates the utilization of path planning algorithms to determine the optimal trajectory among waypoints as multiple goals. Precise estimate of the robot pose (X, Y, Yaw) is demanded by map building so that accurate registration of the local map on the global map is capable of being carried out. In addition, a Kalman Filter is implemented to estimate robot pose reliably by fusing data from the motor encoders, the DGPS, and the digital compass. This map building aims to construct a probabilistic-occupancy-cell-based map. The values for each cell in the map vary over the range [-127, 128]. The initial value is zero, which indicates that the cell is neither occupied nor unoccupied. The value is 128 if one cell is occupied with certainty and -127 if one cell is unoccupied with certainty. The values falling into (-127, 128) express contain level of certainty in between.

III. PSO-BASED MULTI-GOAL VISIT

In real world applications of multi-goal navigation and mapping, multiple goals as a sequence of waypoints and a relative cost for travelling between each goal to each other are provided. The objective is to search a route through all the waypoints in which all waypoints are visited once, and to find the shortest overall tour. In rescue robot application, for instance, the robot starts at one designated waypoint, visits each other waypoint and then ends at the initial waypoint. Traveling Salesman Problem (TSP) is an optimization problem to minimize the travelling distance in a finite number of cities while the cost of travel between each city is known. The classic TSP is employed to deal with the multi-goal visit problem, in which a sequence of goals is visited so that the total planned length of the route is minimized. The objective of this TSP with regard to multi-goal navigation is to search an ordered set of all the waypoints for the autonomous robot to visit at such that the cost is minimized. A list of waypoints and distances, or cost, between each of them is necessary for TSP. Therefore, goals are called waypoints with GPS coordinates of the goals in latitude and longitude. The multi-goal visit problem has practical applications on some fields such as transportation plan issues where deliveries are required as well as fuel cost and time are to be minimized.

In this paper, particle swarm optimization (PSO) is utilized to resolve the TSP for the multi-goal navigation with multiple waypoints ([12]) programmed in MATLAB. PSO algorithm is an approach for optimization on metaphor of social behavior of flocks of birds and swarm of fish by

simultaneously maintaining several candidate solutions in the search space. PSO is an algorithm based on population like genetic algorithms (GAs). The PSO optimizer searches for the optimum solution iteratively after it initiates in a sequence of randomly generated potential solutions. During each iteration of the PSO algorithm, the objective function to be optimized is utilized to evaluate per candidate solution. Maximum or minimum of the objective function is found via fitness landscape with a flying particle that represents each candidate solution. The PSO algorithm is decomposed of three phases that are iteratively executed until stopping criterion is met.

The PSO algorithm for TSP to visit multiple waypoints is summarized as Fig. 1 ([12-13]). The *best* position that a given particle achieves is defined to be *pbest*. The best position achieved so far is tracked by any particle that is called *gbest*.

The particles modify their positions through their velocitie moves or toward *pbest* and *gbest.* A population is initially generated before all particles are evaluated and $pbest_p$ is replaced by x_p, p's position. $Gbest_p$ represents the best position achieved so far by any of the p's neighbors. Finally, the velocities and positions of each particle are updated. The p's velocity, v_p is updated by the p's position, x_p, $pbest_p$ and $gbest_p$. The procedure *compute_velocity*( ) receives three inputs. The behavior of the swarm is acquired by the equations as follows.

$$v_p(t+1) = v_p(t) + \vartheta_1[pbest_p(t) - x_p(t)] + \vartheta_2[gbest_p(t) - x_p(t)],$$

$$x_p(t+1) = x_p(t) + v_p(t+1),$$

where, $v_p(t)$ represents the particle velocity, $x_p(t)$ is the position, at instant t, ϑ_1 and ϑ_2 represent random variables with introduction of uncertainty. $pbest_p(t)$ is the best position the particle achieved up to instant t, $gbest_p(t)$ is the best position that any of p's neighbors has achieved up to instant t,

```
procedure PSO
   Initialize a population of particles
   do
      for each particle p with position xp do
            if (xp is better than pbestp) then
               pbestp ← xp
            end_if
      end_for
      Define gbestp as the best position found so far by
               p's neighbors
      for each particle p do
         vp ← Compute_velocity(xp, pbestp, gbestp)
         xp ← update_ position(xp, vp)
      end_for
while (a stop criterion is not satisfied)
```

Fig.1 PSO algorithm for TSP with multi-waypoints

After execution of this PSO-based TSP algorithm, the minimized total route to connect all the waypoints is obtained. There are 17 waypoints in the workspace that required a service mobile robot to reach per waypoint. The PSO-based TSP algorithm is applied to this application illustrated in Fig. 2. The fitness value over the iterations in terms of the PSO is depicted in Fig.3.

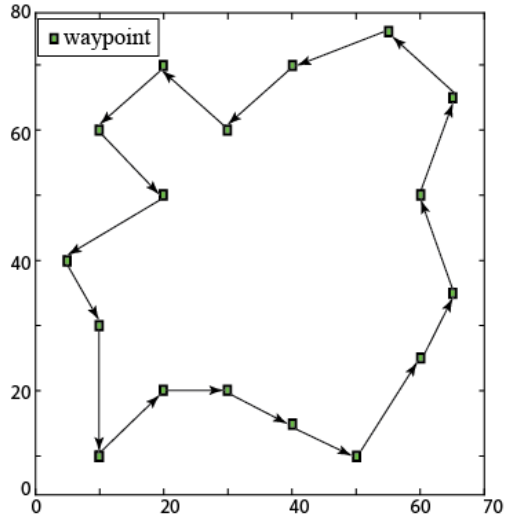


Fig.2 PSO algorithm for TSP applied in a 17-waypoint workspace

IV. Real-Time Concurrent Map Building and Multi-Goal Navigator

Given a set of waypoints as multiple goals and possible starting points, a matching lookup table for waypoint sequencing is created. The D*-Lite algorithm then provides an initial path marked by markers between pairs of goals, and the VFH navigation algorithm drives the robot alone those markers. Path planning is carried out using the D*-Lite algorithm, which provides the best route between waypoints. D*-Lite can work off a partially complete map of the field, and progressively re-plans the optimal route when the map is augmented with new information as the robot explores.

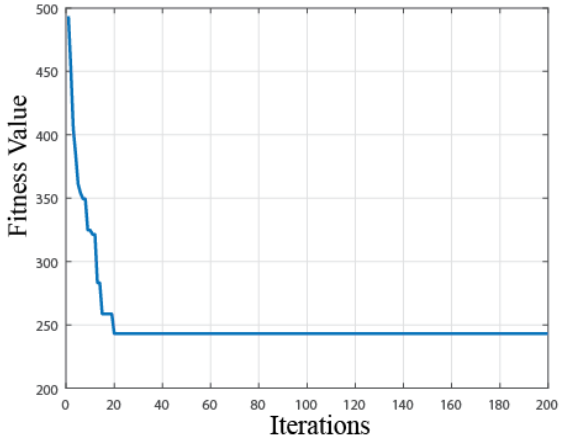


Fig.3 The fitness value over iterations

Although D*-Lite is a very effective algorithm to use for unknown terrain, its integration with a specific navigation task is difficult without a built-in debugging tool. Therefore, we designed a debugging tool which shows the markers path from one waypoint to another as D*-Lite plans. Our program prints out these markers, which update continuously according to the movement of the robot.

A sort of multi-layer software development environment is implemented based on Player/Stage, an open-source, Unix-based robotic simulation software system, in which the proposed algorithms are developed, debugged, tested and simulated. Player provides an interface for a variety of robot and sensor peripherals accessed from local computers via a network connection to the robot whereas Stage™ is a simulation system for a robot and its environment. Since Stage™ can promptly simulate, validate and test a developing algorithm under an off-robot environment, it usually acts as a simulator with the robot, obstacles and sensors (such as LIDAR, camera, digital compass and GPS), to accelerate the development procedure of a new algorithm. Stage™ provides a powerful simulation environment that can be employed to develop and test algorithms in environments similar to those expected in an actual autonomous robot.

These multiple layers on Player/Stage provide modular isolation and enable seamless communication between software and either actual hardware or a hardware simulator possible. The first is the server layer which is occupied by Player, an open-source, Unix-based robotic software system that serves as an interface between robotic algorithm implementations and the robot systems. Specifically, it provides standard interfaces for a typical set of robotic peripherals (LIDAR, cameras, motors, etc.). The second layer is the client layer occupied by user programs written in MATLAB. These layers communicate with Player over a TCP socket to acquire data from sensors and send actuator commands, which Player then passes on to the robot. An actual autonomous robot was developed pictured in Fig. 4 as test-bed for our hybrid system of real-time concurrent multi-goal navigation and map building of an autonomous robot. The robot incorporates six sensors into its compact design as follows: a LIDAR, a DGPS, a digital compass, a camera, and an IMU, each of which is enclosed in a waterproof case and firmly mounted to the robot.

A 270^0 SICK LMS111 LIDAR is configured for the purposes of obstacle detection illustrated in Fig.4. The unit is able to obtain data over a 270° field-of-view with 0.25° resolution, a maximum range of 20 m, and a 25Hz scanning rate. The sensor configuration of our autonomous robot for multi-goal navigation and mapping is illustrated in Figure 3. The autonomous robot driven by D*-Lite global path planning in conjunction with VFH local navigator simulated and tested in a simple scenery with U-shaped and semi-circle obstacles on Player/Stage is depicted in Fig. 4, in which the robot starts at Waypoint A(0, -21) and reaches goal at B(0, 24) with obstacle avoidance.

V. Simulation and Experiments

In this section, the proposed hybrid system for real-time concurrent multi-goal navigation and mapping of an autonomous robot in unknown environments will be validated first on Player/Stage simulator and then on an actual autonomous robot.

In order to validate the effectiveness and efficiency of the proposed hybrid system, the model is applied to simulate such real-time concurrent multi-goal navigation and map building of an autonomous robot under unknown environments in a thirty-goal course.

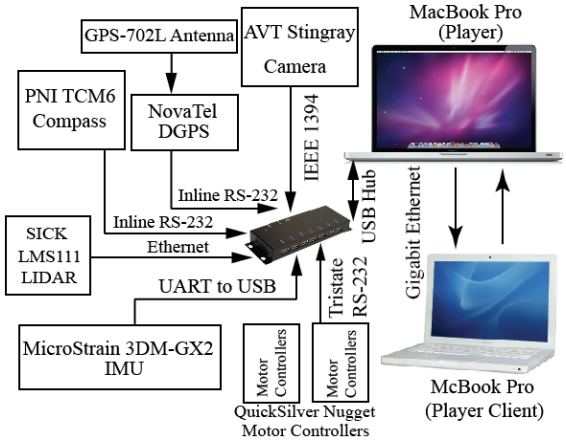
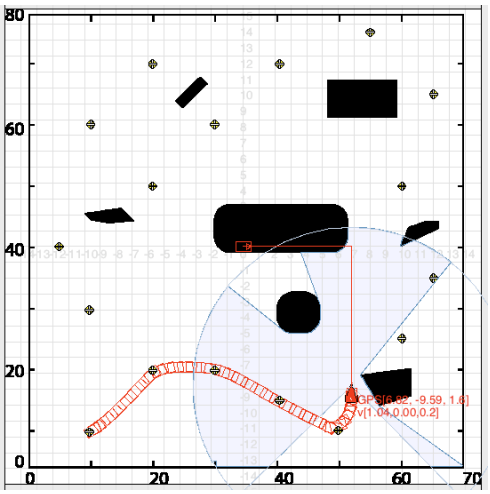


Fig. 4 The sensor configuration of our autonomous robot for multi-goal navigation and mapping



(a)

(b)

Fig 5. Simulation result of the robot in unknown workspace with 17-waypoint by the proposed PSO-based TSP model at the early stage of movement. (a): The workspace with obstacles and 17 goals; (b): Built map.

In this simulation, the robot is navigated to connect 17 waypoints in a free-space to test the PSO-based multi-goal model with mapping at the early stage illustrated in Fig. 5.

The robot is able to traverse from the initial point to plan the shortest route to visit the 17 waypoint while the robot builds the map with 270 LIDAR. The rout of multi-goal by the robot is shown in Fig. 5(a) whereas the map built is illustrated in Fig. 5(b) at the end of the travel of the robot. In this simulation, the robot is guided in an *unknown* environment populated with obstacles with 17 waypoints as multiple goals depicted in Fig. 5 , which shows that the robot traverses from starting waypoint to connect 17 waypoints obtained by GPS coordinates. The built map while the robot moves in the unknown environment with 270º LIDAR scan is illustrated in Fig. 5(a) while the robot moves on the road from waypoint from waypoint. The yellow portions are detected obstacles in Fig. 5(b).

In accordance with the planed trajectory, the GPS coordinates of the targets in latitude and longitude in sequence of the goals is dispatched to the D*-Lite based global path planner. As described in the previous sections, from one goal to another goal, the VFH based local navigator is utilized to navigate the robot by following pre-placed markers. Starting from the initial point, the autonomous robot is capable of travelling to the sequence of goals based on the proposed multi-goal navigation system and experiment was successfully accomplished.

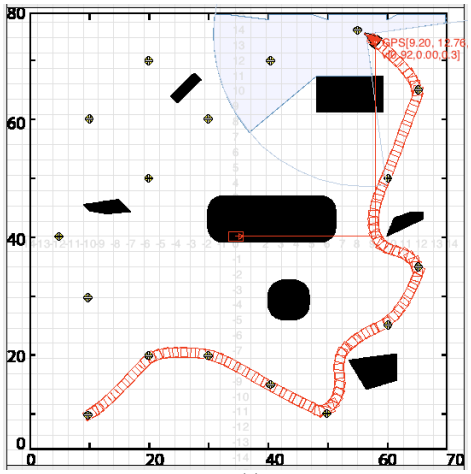
In the experimental test field, the robot is required to autonomously travel from a predefined starting point to a number of target destinations as waypoints and move back to home base, in which the waypoints are given only by the GPS coordinates of the targets in latitude and longitude.

After the route to visit these fifteen waypoints is generated by the proposed PSO-based TSP algorithm, the GPS coordinates of the targets in latitude and longitude in sequence of the goals are transmitted to the global path planner in Fig. 6. Once a path from D* Lite has been obtained that was early generated by the PSO-based TSP algorithm, a number of points along it are extracted so as to use the VFH algorithm between two goals. These points as markers are converted into GPS coordinates and presented to VFH as consecutive points. VFH then generates motion commands which are transmitted to the drive controllers to move the robot towards these intermediate waypoints. The planned route and map built of the robot in unknown workspace with 17-waypoint at the middle stage of movement are illustrated in Fig. 6(a) and Fig. 6(b), respectively.

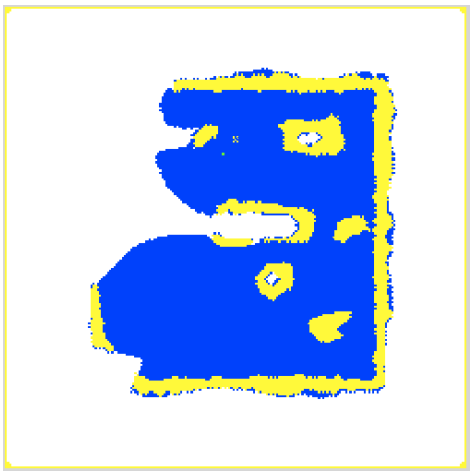
Once the robot approaches to an intermediate waypoint, the next intermediate goal along the desired path is regarded to be achieved and substituted. The planned route and map built of the robot in unknown workspace with 17-waypoint while it has traverses the most region are illustrated in Fig. 7(a) and Fig. 7(b), respectively.

In Fig. 8, the autonomous robot returns the starting point after it traverses from starting waypoint to every goal by planning a collision-free trajectory to connect these thirty goals with minimized overall distance.

Once the robot approaches the initial waypoint, the all the waypoints are visited once. The planned route and map built of the robot in unknown workspace with 17-waypoint at the end are illustrated in Fig. 8(a) and Fig. 8(b), respectively.



(a)

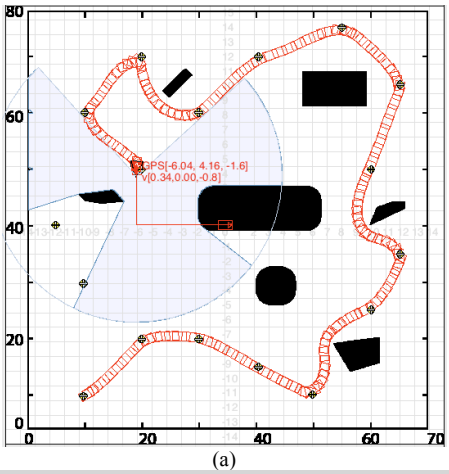


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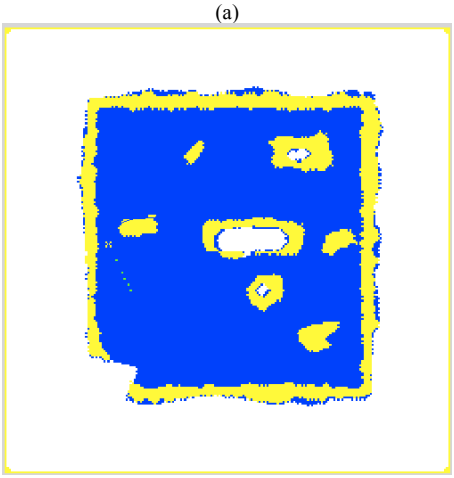
Fig 6. Simulation result of the robot in unknown workspace with 17-waypoint at the middle stage of movement by the proposed PSO-based TSP model. (a): The workspace with obstacles and 17 goals; (b): Built map.

VI. Conclusion

A solution by hybrid algorithms was developed in this paper for real-time map building and navigation for **multiple goals** purpose. In this paper, an alternate approach, the D*-Lite algorithm associated with a local LIDAR-based navigation methodology was developed for multiple goals. The D*-Lite path planning algorithm was used to provide VFH with intermediate goals. The multi-goal route was calculated and planned by the proposed PSO-based TSP strategy. Results from simulation studies and experiments demonstrated the benefits of the local navigator in conjunction with a path planner to reach multiple goals with minimized total distance.

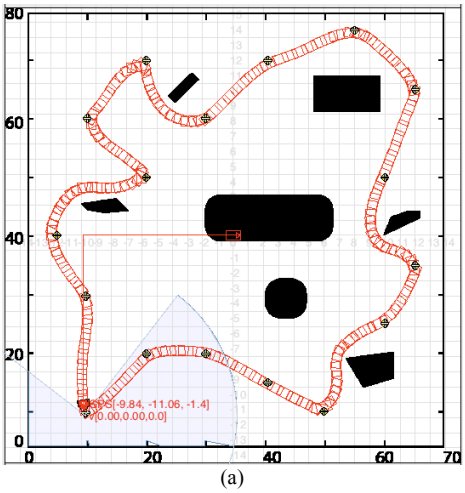


(a)

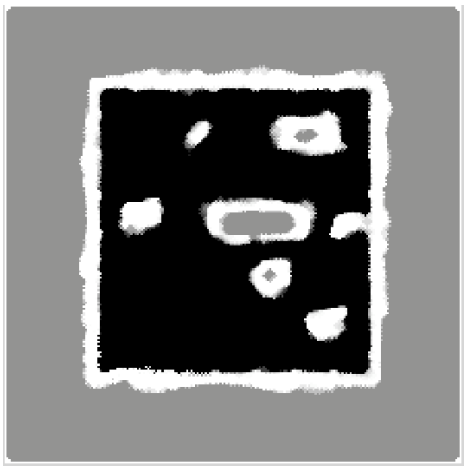


(b)

Fig. 7 The route planned by the proposed PSO-based TSP model during the travel of the robot (a): The workspace with obstacles and 17 goals; (b): Built map.



(a)



(b)

Fig 8. Simulation result of the robot in unknown workspace with 17-waypoint by the proposed PSO-based TSP model at the end. (a): The workspace with obstacles and 17 goals; (b): Built map.

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